

Vocational Training Authority of Sri Lanka

NetPlans Selling System

Web Programming Project-2025
K72T001M07

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of NetPlans, a PHP and MySQL web application that lets customers browse and purchase pre-designed network diagrams/plans (for hospitals, schools, hotels, enterprises, and home labs). The purpose of this document is to provide a clear reference of the system's intended behaviour for developers, evaluators, and maintainers, based on an analysis of the existing project source code.

1.2 Scope

NetPlans is a public-facing website where visitors can browse network plan templates by category, view plan details, register/login, purchase a plan (checkout), request a customized plan, and contact the site owner. A separate administrator panel allows staff to upload new plans, manage client orders and their statuses, manage (block/unblock) client accounts, and review contact messages.

The system does not currently support online payment gateway integration, file/PDF delivery of purchased plans, or password-reset via email. These are noted as potential future enhancements in Section 2.6.

1.3 Definitions, Acronyms and Abbreviations

Term	Description
SRS	Software Requirements Specification
PHP	Hypertext Preprocessor (server-side scripting language)
PDO	PHP Data Objects (database access layer)
Client	A registered visitor who can browse and purchase plans
Admin	A staff user with access to the administration panel

1.4 References

- PHP Manual (php.net)
- MySQL 8.0 Reference Manual
- PHP PDO Documentation
- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications

1.5 Overview

The remainder of this document is organized as follows: Section 2 gives an overall description of the product; Section 3 details functional requirements by feature; Section 4 describes external interface requirements; Section 5 specifies non-functional requirements; Section 6 lists primary use cases with a diagram; and Section 7 provides the inferred database schema with an ER diagram.

2. Overall Description

2.1 Product Perspective

NetPlans is a self-contained web application following a classic multi-page PHP architecture. Each page connects directly to a MySQL database via PDO to render dynamic content and process form submissions. The application has two access areas: a public/client-facing site and a separate admin area under the /admin directory, each with its own session-based authentication.

2.2 Product Functions

At a high level, the system provides the following functions:

- Client registration and login with securely hashed passwords
- Browsing network plans by category on the gallery page and homepage
- Viewing detailed information for an individual plan
- Purchasing a plan (checkout), creating an order tied to the logged-in client
- Viewing a client's own order history on their dashboard
- Submitting a plan customization request or a general contact message
- Administrator login separate from client login
- Administrator dashboard showing totals for active plans, orders, clients, and revenue
- Administrator plan upload (title, category, description, price, image)
- Administrator order status management (pending/confirmed/cancelled)
- Administrator client management (block/unblock accounts)
- Administrator review of submitted contact messages

2.3 User Classes and Characteristics

User Class	Description
Guest (Visitor)	An unauthenticated visitor who can view the homepage, about page, gallery, and plan details, and submit contact/customization forms.
Client	A registered user (role = client) who can log in, purchase plans, and view their own order history.
Administrator	A user with role = admin who can log into the separate admin panel to manage plans, orders, clients, and contact messages.

2.4 Operating Environment

- Server: Any web server supporting PHP (e.g. Apache via XAMPP) with PHP 7.4+ recommended
- Database: MySQL/MariaDB (database name: network_plans_db)
- Client: Any modern web browser with JavaScript/CSS support
- PHP Extensions: PDO and pdo_mysql for database access

2.5 Design and Implementation Constraints

- Database connection details (host, database name, username, blank password) are hard-coded and duplicated across several files (includes/db.php and individually inside each admin page) rather than centralized in a single configuration file.
- Uploaded plan images are stored directly on the server's filesystem (an /uploads directory) with filenames based on upload time, and the file path is stored in the plans table.
- Session-based authentication is used separately for clients (user_id) and administrators (admin_id), with no shared single sign-on.

2.6 Assumptions and Dependencies

- It is assumed the MySQL database and required tables (users, plans, orders, contacts) already exist and are reachable before the application is used.
- It is assumed a single administrator account (or a small trusted set) manages the admin panel; there is no role hierarchy beyond client/admin/blocked.
- Future enhancements could include real payment gateway integration, automatic delivery of purchased plan files, email notifications, and password recovery.

3. System Features (Functional Requirements)

3.1 User Registration & Authentication

Description: Allows a visitor to create a client account and subsequently log in or out.

ID	Requirement
FR-1.1	The system shall allow a visitor to register with name, email, password and password confirmation.
FR-1.2	The system shall reject registration if the password and confirmation do not match.
FR-1.3	The system shall securely hash passwords (bcrypt) before storing them, and shall reject registration with a duplicate email address.
FR-1.4	The system shall allow a registered client to log in with email and password, verifying the password against the stored hash.
FR-1.5	The system shall create a session for a successfully authenticated client and redirect them to their dashboard.
FR-1.6	The system shall allow a logged-in client to log out, ending their session.

3.2 Plan Browsing & Details

Description: Allows visitors to discover and inspect network plans.

ID	Requirement
FR-2.1	The system shall display up to six active plans on the homepage.
FR-2.2	The system shall display all active plans grouped by category (Hospital, School, Hotel, Enterprise, Home Lab) on the gallery page.
FR-2.3	The system shall allow a visitor to open a plan's dedicated details page and view its title, category, description, price, and image.
FR-2.4	The system shall display an appropriate message if a requested plan ID does not exist.

3.3 Checkout & Orders

Description: Allows an authenticated client to purchase a plan and review their order history.

ID	Requirement
FR-3.1	The system shall require a client to be logged in before accessing the checkout page, redirecting to login otherwise.
FR-3.2	The system shall display the selected plan's details on the checkout page for confirmation before purchase.
FR-3.3	The system shall create an order record (linked to the client and the plan, with total price and status 'confirmed') when checkout is submitted.
FR-3.4	The system shall confirm the order and redirect the client to their dashboard after checkout.
FR-3.5	The system shall display a client's own order history (plan title, price, status, date) on their dashboard, requiring login.

3.4 Customization & Contact Requests

Description: Allows visitors and clients to send messages to the site owner.

ID	Requirement
FR-4.1	The system shall provide a customization request form (name, email, requested changes) that stores the message as a contact record with subject 'Customization Request'.
FR-4.2	The system shall provide a general contact form (name, email, subject, message) that stores the message as a contact record.

3.5 Administrator Authentication

Description: Provides a separate, restricted login for administrative staff.

ID	Requirement
FR-5.1	The system shall provide a separate admin login page that authenticates only users whose role is 'admin'.
FR-5.2	The system shall create a distinct admin session upon successful admin login and redirect to the admin dashboard.
FR-5.3	The system shall restrict all admin pages (dashboard, manage plans/orders/users/contacts) to users with an active admin session, redirecting to the admin login page otherwise.

3.6 Administrator Dashboard & Plan Management

Description: Gives the administrator an overview and the ability to add new plans.

ID	Requirement
FR-6.1	The system shall display, on the admin dashboard, the count of active plans, total orders, total clients, and total confirmed revenue.
FR-6.2	The system shall allow the administrator to upload a new plan with title, category, description, price, and an image file.
FR-6.3	The system shall store the uploaded image on the server and save its file path in the plan record, marking the new plan as active by default.
FR-6.4	The system shall display an error message if no image is provided or the image upload fails.

3.7 Administrator Order & User Management

Description: Allows the administrator to manage client orders and accounts.

ID	Requirement
FR-7.1	The system shall display all orders with client name, email, plan title, price, status, and date, ordered by most recent.
FR-7.2	The system shall allow the administrator to update an order's status (e.g. pending, confirmed, cancelled).
FR-7.3	The system shall display all client accounts (excluding admins) with their registration date and current status (active/blocked).
FR-7.4	The system shall allow the administrator to block an active client or unblock a previously blocked client.

3.8 Administrator Contact Message Review

Description: Allows the administrator to review messages submitted via the contact and customization forms.

ID	Requirement
FR-8.1	The system shall display all submitted contact/customization messages with sender name, email, subject, message, and date, most recent first.

Implementation note: the login page currently does not check whether a client's role has been set to 'blocked', meaning a blocked client may still be able to log in; this is flagged here as a gap for review rather than an intended requirement.

4. External Interface Requirements

4.1 User Interfaces

- The public site provides a graphical web interface (HTML/CSS) consisting of: Home, About, Gallery, Plan Details, Customize, Contact, Register, Login, Checkout, and Dashboard pages, with a shared header/footer.
- The admin panel provides a separate set of pages: Admin Login, Admin Dashboard, Manage Orders, Manage Users, Manage Contacts, and Upload Plan.
- Navigation is provided via a top navigation bar; login state (client name, or login/register links) is reflected conditionally in the header.

4.2 Hardware Interfaces

No special hardware is required. The application runs on any standard web server hardware and is accessed by clients through any device with a modern web browser.

4.3 Software Interfaces

Component	Purpose
PHP (with PDO, pdo_mysql)	Server-side application logic and database access
MySQL / MariaDB	Stores users, plans, orders, and contacts data
Web Server (e.g. Apache)	Serves the PHP application over HTTP

4.4 Communication Interfaces

The application communicates with the MySQL database over a local database connection (default: localhost). The client browser communicates with the web server over standard HTTP.

5. Non-Functional Requirements

5.1 Performance Requirements

- Pages that query the database (gallery, plan details, dashboard, admin lists) should render within a few seconds under normal load and typical data volumes.

5.2 Security Requirements

- Client passwords are hashed using bcrypt (password_hash/password_verify), which is good practice and should be preserved in any future changes.
- Database credentials (MySQL root user with a blank password) are hard-coded in multiple files; centralizing and securing these credentials is recommended before production deployment.
- SQL queries use parameterized/prepared statements (PDO prepare/execute) throughout, which mitigates SQL injection risk.
- The client login process does not currently reject accounts with role 'blocked'; this should be addressed so blocked clients cannot continue to log in.
- Uploaded plan images are saved using a time-based filename, but file type/content is not validated, which is a risk that should be reviewed (e.g. restrict to image MIME types/extensions).

5.3 Usability Requirements

- The interface shall provide clear success or error messages for registration, login, checkout, and form submissions.
- Category-based browsing on the gallery page shall make it straightforward for visitors to find relevant plans.

5.4 Reliability and Availability

- The system shall handle database connection failures by displaying a clear error message rather than an unhandled crash.
- As a single-server web application, availability depends on the web server and MySQL database remaining online.

5.5 Maintainability

- Shared layout is partially factored into includes/header.php and includes/footer.php, but the database connection code is duplicated across several files instead of being reused from includes/db.php; consolidating this would ease future maintenance.

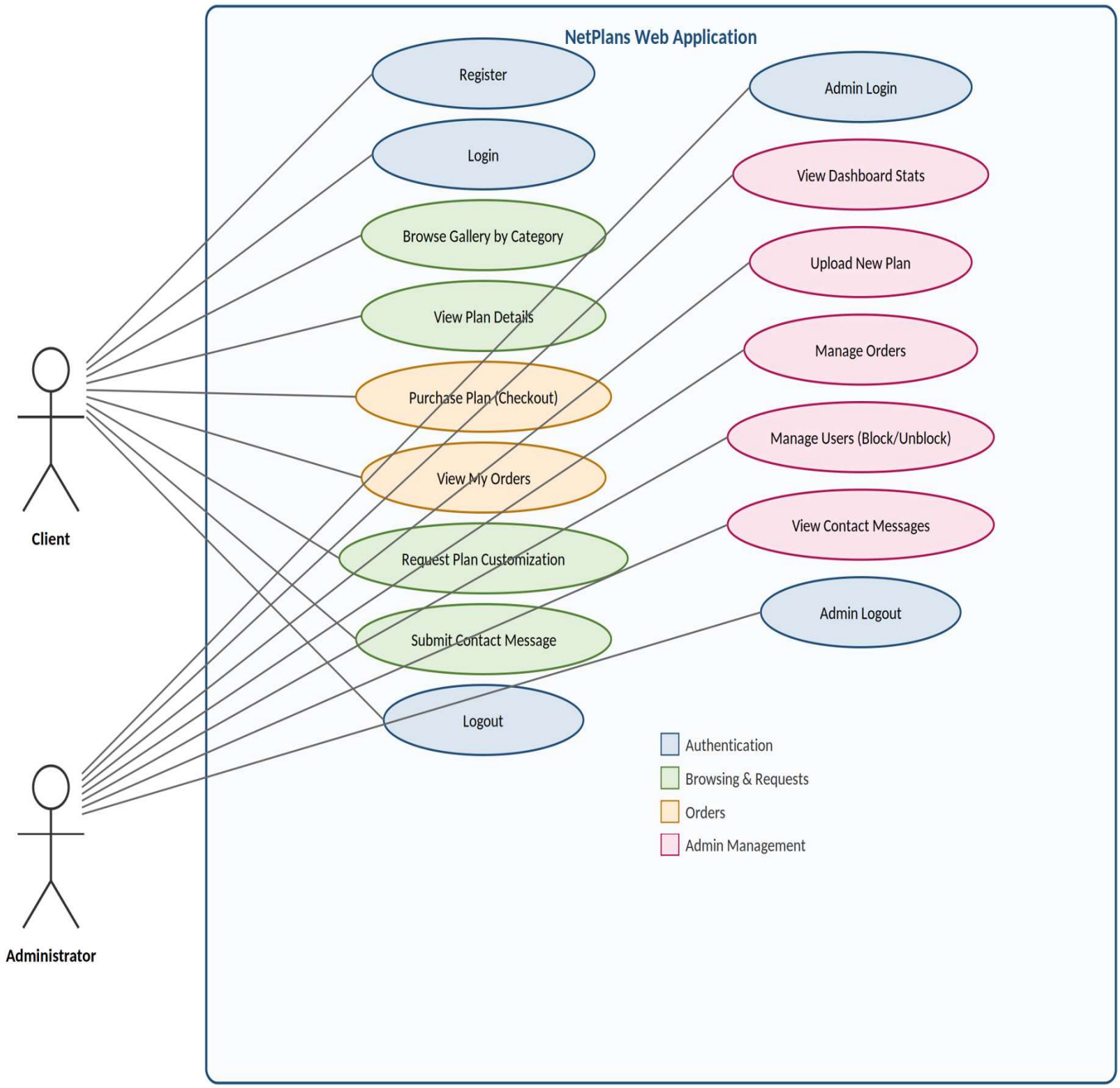
5.6 Portability

As a standard PHP/MySQL application, the system can run on any hosting environment that supports PHP with the PDO MySQL extension, without modification to the source code.

6. Primary Use Cases

The diagram below shows the two actors (Client and Administrator) and the use cases supported by the system, grouped by functional area.

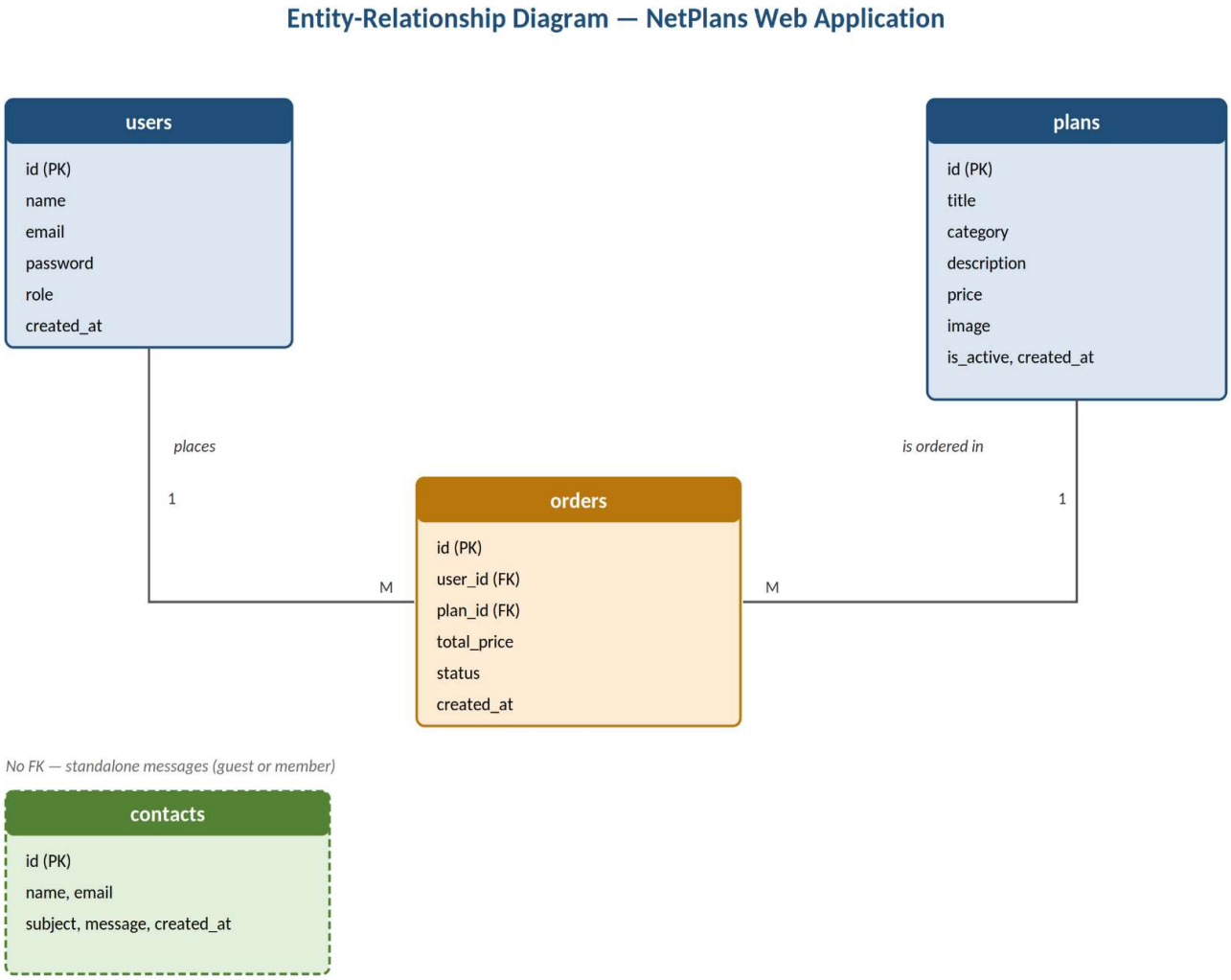
Use Case Diagram — NetPlans Web Application



Use Case	Actor	Description
Register	Client	Visitor creates a new client account.
Login	Client	Client authenticates to access checkout and dashboard.
Browse Gallery by Category	Client	Visitor views plans grouped by category.
View Plan Details	Client	Visitor views full details of a single plan.
Purchase Plan (Checkout)	Client	Logged-in client buys a plan, creating an order.
View My Orders	Client	Client views their own order history on the dashboard.
Request Plan Customization	Client	Visitor/client requests a customized network plan.
Submit Contact Message	Client	Visitor/client sends a general enquiry.
Logout	Client	Client ends their session.
Admin Login	Administrator	Administrator authenticates to access the admin panel.
View Dashboard Stats	Administrator	Administrator views summary counts and revenue.
Upload New Plan	Administrator	Administrator adds a new plan with an image.
Manage Orders	Administrator	Administrator views orders and updates their status.
Manage Users (Block/Unblock)	Administrator	Administrator blocks or unblocks client accounts.
View Contact Messages	Administrator	Administrator reviews submitted messages.
Admin Logout	Administrator	Administrator ends their session.

7. Appendix: Inferred Database Schema

The following entity-relationship diagram and tables were inferred from the SQL statements used throughout the source code (database: network_plans_db). A client can place many orders, and a plan can appear in many orders; contact messages are standalone and are not linked by a foreign key.



7.1 users

Column	Notes
id	Primary key / unique identifier of a user
name	User's full name
email	Unique login email address
password	Bcrypt-hashed password
role	'client', 'admin', or 'blocked'
created_at	Registration timestamp

7.2 plans

Column	Notes
id	Primary key / unique identifier of a plan
title	Plan title
category	Hospital, School, Hotel, Enterprise, or Home Lab
description	Plan description text
price	Plan price
image	Server file path of the uploaded plan image
is_active	Whether the plan is currently shown to visitors
created_at	Upload timestamp

7.3 orders

Column	Notes
id	Primary key / unique identifier of an order
user_id	References users.id
plan_id	References plans.id
total_price	Price recorded at time of purchase
status	'pending', 'confirmed', or 'cancelled'
created_at	Order timestamp

7.4 contacts

Column	Notes
id	Primary key / unique identifier of a message
name	Sender's name
email	Sender's email
subject	Message subject (e.g. 'Customization Request')
message	Message body
created_at	Submission timestamp

Note: Column names and types are inferred from application queries; exact SQL data types and constraints should be confirmed against the live database schema.